

PetReunion: A Preliminary Interview Study to Understand User' Opinions on Using Mixed Reality to Support Pet Bereavement

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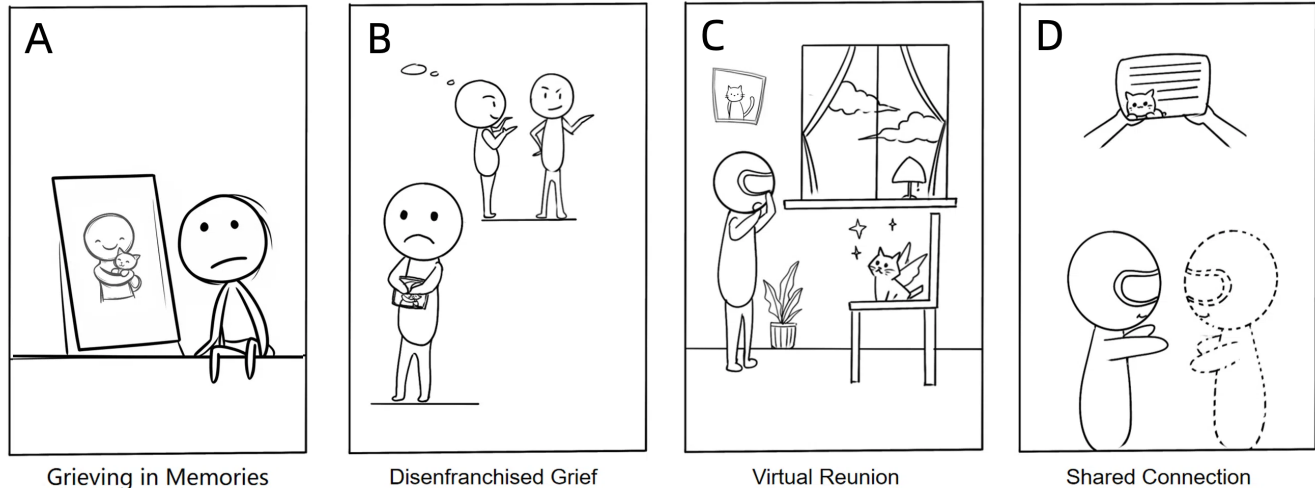


Figure 1: This figure shows a conceptual illustration of the core idea of this paper, showcasing the role of mixed reality in facilitating connection and support during pet bereavement: (A, B) People experiencing pet loss often face deep sadness and disenfranchised grief from those around them. (C, D) Our conceptual MR system aims to provide comfort by enabling virtual re-connections and fostering empathy through shared experiences.

Abstract

Pet ownership brings emotional well-being. However, when a pet dies, the emotional bond can result in grief that is not socially acknowledged. This disenfranchised grief often leaves bereaved pet owners without emotional support. While human-computer interaction (HCI) research has explored technologies for human bereavement, pet bereavement remains underexplored. Mixed reality (MR) can help bridge this gap. It allows users to create immersive and personalized spaces to reconnect with deceased pets. This paper presents findings from a preliminary study involving 10 participants who experienced pet loss within the past three years, investigating their emotional needs and coping behaviors. The study shows the limitations of current approaches and highlights the benefits of

interactive, multisensory experiences. Based on these insights, this paper provides design considerations for MR systems tailored to reduce owner sadness and support pet bereavement. The long-term research goal is to develop immersive, user-centered systems that provide meaningful and long-term emotional support.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Mixed / virtual reality**.

Keywords

pet bereavement, mixed reality, emotional support, continuing bonds, virtual pet interactions

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1 Introduction

The bond between humans and animals has deepened, with pets increasingly regarded as family members, providing emotional and psychological benefits such as reduced stress and increased feelings of trust [16]. However, these relationships inevitably involve the challenge of pet loss, which can evoke grief comparable to losing a human loved one. Owners often experience sadness, guilt, or anger, especially when a pet's death is sudden or unexpected [34, 46]. This grief can persist for months or years, and its intensity is often socially unacknowledged, leading to disenfranchised grief [5, 35]. Unlike human bereavement, which is supported by communal rituals, pet loss often lacks socially sanctioned outlets, exacerbating emotional isolation [6].

To cope with the loss, many pet owners turn to Continuing Bonds (CB) practices such as preserving possessions, viewing photos, or creating memorials [10, 28]. However, these static and passive methods do not fully capture the evolving nature of grief or engage multiple senses, limiting their emotional impact. Mixed Reality (MR) offers a novel way to address these limitations by combining real-world environments with interactive 3D-rendered elements. MR enables immersive and personalized experiences, allowing users to actively engage with virtual representations of pets, fostering emotional expression and memory recall through multisensory interaction (Figure 1). This dynamic approach acknowledges the complexity of grief, providing a healing space that evolves with the user's emotional journey.

This paper presents findings from a preliminary study with 10 participants who experienced pet loss within the past three years, exploring their emotional experiences and coping mechanisms. These insights inform the design of MR systems tailored for pet bereavement, emphasizing the need for personalized, engaging, and emotionally sensitive support. By addressing the emotional and social challenges of pet bereavement, this work contributes to the growing field of MR applications in grief support. It provides design insights for creating immersive systems that facilitate continuing bonds, laying the foundation for future technological interventions to support grieving pet owners.

2 Related Work

2.1 Continuing Bonds Theory

CB theory has been widely adopted in bereavement research to understand how people maintain meaningful relationships with the deceased, emphasizing emotional adjustment and healing through ongoing relationships. Within HCI, CB theory inspires systems that blend physical and digital elements. For instance, *Be.side* and *ReFind* facilitate personalized rituals and dynamic memory curation, offering tools for reflection and ongoing engagement [19, 45]. Systems like *ThanatoFenestra* and *SenseCenser* combine sensory technologies with traditional memorial practices, highlighting the role of embodied interactions in maintaining these bonds [42, 43]. Interactive technologies extend CB theory into immersive environments. *SenseVase* integrates VR with floral tributes, and Japan's automated columbaria systems blend physical and digital interactions, enabling engagement with cremated remains in technologically

mediated ways [40, 41]. These works illustrate the potential of interactive systems to reshape grief by fostering adaptive, meaningful engagements with memories and emotions.

2.2 Pet Bereavement

The pet-human bond has gained recognition for its emotional significance, and interactive technologies enhance these connections. Examples include *LevPet*, a magnetic levitating pet enabling dynamic engagement, and *PetPresence*, which integrates real-world pet activities into virtual environments [38, 47]. Liang et al. further advanced virtual pet interactions with scene-aware behavior synthesis for MR, creating more natural, interactive behaviors [21].

Pet loss often leads to grief as intense as that experienced after human loss, affecting emotional well-being and daily functioning [26, 34]. Grief can persist for months or years, with some individuals reporting ongoing symptoms a year after their pet's death [13, 26, 46]. Owners adopt coping mechanisms like preserving belongings, looking at photos, or writing letters to maintain emotional connections [10, 28]. Writing eulogies also helps process grief and offer comfort [12, 22]. However, pet bereavement often involves disenfranchised grief, as society tends to undervalue pet loss. This lack of recognition prevents grieving owners from receiving the emotional support they need, leading to feelings of isolation [5, 35]. In contrast, bereavement rituals for humans, such as funerals, provide shared opportunities for mourning and healing [7, 29]. Our work builds on these foundations, along with prior research revealing the potential of immersive technologies in bereavement [30, 40, 48], by exploring how MR can support pet bereavement through immersive, multisensory experiences. By integrating virtual pet interactions and adaptive designs, MR systems can create dynamic spaces for individuals to process emotions and maintain meaningful connections.

2.3 Immersive Technologies in Grief Support

Immersive technologies like Virtual Reality (VR) are increasingly used in mental health interventions due to their ability to create controlled, immersive environments that foster presence and allow users to engage with emotionally impactful scenarios. VR systems have shown potential in grief support by recreating virtual encounters with deceased loved ones, facilitating emotional closure and comfort. For instance, systems that simulate conversations or interactions with the deceased help users process emotions in deeply personal ways [20, 30, 39]. Narrative-driven experiences like *Road to Acceptance* guide users through stages of grief, enabling emotional reflection and catharsis [1, 17]. These applications demonstrate VR's capacity to foster empathy and catharsis. However, VR's challenges—such as motion sickness, user discomfort, and the isolating nature of headsets—limit its suitability for populations with sensory sensitivities or Autism Spectrum Disorder (ASD) [2, 18]. Fully immersive environments may also disconnect users from their physical surroundings, reducing applicability in contexts requiring real-world grounding [32, 37]. MR overcomes many of these limitations by integrating virtual elements into real-world environments. Unlike VR, MR blends physical and virtual objects, maintaining a connection to the present environment while fostering engagement, a critical feature for grief support where isolation is a risk [24, 27]. Compared to mobile Augmented Reality,

which often relies on 2D screens and limited interactivity, MR leverages head-mounted displays to offer a broader field of view and hands-free, multisensory engagement (e.g., spatial audio, haptic feedback potential), enhancing the emotional depth of virtual pet interactions. MR also supports collaborative experiences, enabling therapists or family members to interact with the bereaved in real time, creating an empathetic and supportive space [44].

Building on these foundations, our work develops an MR system for pet bereavement, integrating multisensory elements and personalized virtual pet interactions. This approach provides a dynamic, evolving space for grieving individuals to process emotions and maintain continuing bonds with their lost companions.

3 Preliminary Study

3.1 Interviews and Participants

To gain an in-depth understanding of individual pet bereavement experiences—including coping mechanisms and perceptions of technology in grief support—we used a qualitative research design. Semi-structured interviews were conducted to capture detailed narratives, focusing on participants' experiences with pet loss, coping mechanisms, attitudes toward MR for grief support, preferences for technological features, and concerns about its use. We used only this method because bereavement research, as investigating in previous research[19, 43], is a universal yet unique experience, varies widely across individuals. Unlike quantitative methods that generalize, it let us deeply explore each participant's distinct grief. Interviews were conducted via online platforms (Zoom, WeChat, and WhatsApp) based on participant preferences, with the option to withdraw at any time if they felt distressed. The semi-structured interviews were conducted individually with one interviewer per participant over 30–45 minutes, allowing for flexibility to explore participants' emotional narratives while maintaining consistency. Example questions included: 'Can you describe your emotional experience after losing your pet?' and 'How do you feel about using mixed reality to revisit memories of your pet?' Interviews were transcribed verbatim, and analyzed using thematic analysis[3].

We recruited 10 participants aged 21 to 44 years (Mean = 29.1), including six females and four males from diverse geographic locations such as the United States, Canada, the United Kingdom, China, and Morocco. Participants had lost either cats or dogs within the past three years due to causes including old age, illness, accidents, or sudden health issues. This broader timeline allowed us to explore both acute and prolonged emotional responses. For instance, participants' pets included dogs that passed due to old age or illness (e.g., arthritis, dementia, and cancer) and cats lost to conditions like kidney disease and accidents. Bereavement periods ranged from 4 to 36 months. Five participants had prior exposure to VR or MR technology, while the others were unfamiliar. To ensure inclusivity, demonstration videos which showcasing a simple MR scenario (e.g., a virtual object interacting with a real-world environment) were provided to those unfamiliar with MR to give them a basic understanding of its capabilities and potential applications in grief support. This diverse participant pool enabled us to capture a wide range of perspectives and emotional experiences, enriching the study's findings.

3.2 Findings

Our findings are organized into three main themes that emerged from the analysis of participant interviews: (1) *Emotional Landscape of Grief*, (2) *Coping Mechanisms and Practices*, and (3) *Perceptions of Technology in Grief Support*. Each theme reflects the participants' experiences and attitudes towards bereavement and their openness to novel technological interventions.

3.2.1 Emotional Impact of Pet Loss. Participants described a complex and deeply personal emotional journey following the loss of their pets. The emotions experienced were multifaceted and evolved over time, with recurring patterns observed across participants.

Various Grief Journeys Participants described deeply personal and complex emotional journeys after losing their pets. The circumstances of a pet's death significantly shaped their grief. Sudden losses often led to shock and self-blame. Participant I, whose cat passed due to a stress reaction, shared, *"At first, I couldn't believe it happened. I felt numb, but after handling her funeral, the sadness came flooding in."* Conversely, participants who had time to prepare for their pet's death, such as due to old age or illness, experienced anticipatory grief. Participant H reflected, *"From the time he got sick to his passing, I was constantly sad and anxious. Every day, I worried about his condition and feared losing him, but I was also mentally prepared. When he finally passed, my sadness peaked."*

Many participants equated pet loss to losing a family member. Participant C shared, *"Lottie was not just a dog; she was part of my family."* Grief was described as unpredictable and non-linear, with emotions fluctuating over time. Participant G, whose cat passed three years prior, explained, *"Even small reminders, like her photos or old toys, bring back waves of longing. It feels like she never really left."* Triggers such as photos, toys, or routines evoked bittersweet feelings, bringing both comfort and sorrow. These accounts illustrate the profound, enduring emotional impact of pet loss and the deep bond between humans and their pets.

Disenfranchised Grief and Isolation Disenfranchised grief [5, 35], where pet loss is socially undervalued, was frequently mentioned, leading to emotional isolation. Participant E shared, *"When my dog died in a car accident, people around me said, 'It's just a dog,' as if my grief wasn't real. It made me question if I was overreacting, but the pain was as deep as losing a family member."* Cultural norms often compounded these feelings. Participant H reflected, *"In our culture, pets are never seen as equals to family members. Expressing this kind of emotion is sometimes viewed as excessive or immature, which made me feel suppressed."* She added, *"My parents didn't fully understand. My father even blamed me for being moody at home, saying I shouldn't let the cat's death affect the family atmosphere. That made me feel lonely and misunderstood."* Participants also highlighted the lack of formal mourning rituals for pets. Participant F noted, *"If a person dies, there are ceremonies and gatherings where people support each other. But I do this alone for my dog."* These accounts underscore the emotional challenges posed by disenfranchised grief and the societal barriers that leave grieving individuals feeling unsupported and isolated.

3.2.2 Coping Mechanisms and Continuing Bonds. Participants adopted a variety of strategies to cope with the loss of their

pets, reflecting both shared and individual approaches to grief. These strategies included preserving tangible keepsakes, engaging in rituals, and managing emotional responses to memories.

Keepsakes, Tangible Memories, and Rituals Participants preserved keepsakes and engaged in rituals to honor their pets and maintain an emotional bond. Tangible reminders like belongings, ashes, or personal projects served as meaningful connections. Participant C shared, *"I kept Lottie's favorite collar and leash. It's a tangible reminder of her presence and all the memories we shared."* Similarly, Participant H displayed her pet's ashes and fur as keepsakes at home, saying, *"These are at the entrance of my home, so I see them every time I come and go. It feels like he's still with me."* Others created personalized tributes, such as Participant F, who compiled a scrapbook, explaining, *"It helps me relive the happy times and keeps her spirit alive in my heart."* Keepsakes often provided comfort and a physical way to reflect on shared experiences. Symbolic rituals also played a significant role in maintaining these emotional connections. Participant F described lighting a candle on her dog's passing anniversary, saying, *"It's my way of remembering her and reflecting on the love she gave me. It brings me peace."* Participant B celebrated her dog's birthday posthumously, while some participants incorporated routines from their pet's life, such as walking familiar routes, finding these activities meaningful and reflective.

Active Remembrance and Avoidance of Triggers Participants expressed different attitudes toward revisiting memories of their pets. Some found comfort and joy in reflecting on positive moments. Participant G shared, *"Sometimes I take out her old toys or watch her videos. It makes me smile and remember the good times."* However, not all participants found it easy to engage with such mementos. For others, these memories could trigger intense feelings of sadness and loss. Participant H explained, *"I avoid looking at her photos too often because it brings back too much sadness. I miss her so much that it's hard to handle."* Participant J expressed a similar feeling. These contrasting approaches highlight the varied ways individuals cope with grief—some actively engage with memories as part of their healing, while others choose to distance themselves to manage emotional distress.

Social Practices and Community Support In contrast to the isolation they experienced locally, participants found solace in online communities. Social platforms provided validation and a sense of belonging. Participant I shared, *"I posted a letter I wrote to my cat on Little Red Book and connected with others who had similar experiences. It helped me feel understood and less alone."* She also mentioned being inspired by another user's story of her cat's reincarnation: *"It encouraged me to believe that one day I might see my cat again."* Similarly, Participant F described joining an online grief community, saying, *"It felt like a safe space where I didn't have to explain why I was so devastated."* These shared experiences filled the gap left by societal acknowledgment. By engaging in conversations with others who had faced similar losses, participants were able to validate their emotions and find comfort in shared experiences, which helped mitigate the isolation often associated with grief.

3.2.3 Perceptions of Technology in Grief Support. Participants shared diverse attitudes toward using MR to remember their pets. Discussions revealed varying levels of interest, concerns, and interaction preferences.

Interest in Virtual Reunions For some participants, VR/MR offered an appealing way to reconnect with their pets, particularly after sudden losses. Participant I, who lost her cat unexpectedly, shared, *"If I could see my cat again, even virtually, and talk or play with her, it would give me some closure. Her death was so sudden that I never got to say goodbye."* Others expressed interest in revisiting cherished moments or creating a sense of closure. Participant D mentioned, *"Watching a virtual representation of my pet could provide an opportunity to relive cherished memories."* Similarly, Participant A envisioned a virtual park where she could walk her dog again, saying, *"It would be wonderful to revisit those happy moments and feel her presence again, even if it's not real."* For these participants, VR/MR provided a way to maintain continuing bonds and process their grief.

Concerns About Emotional Dependency While some embraced the idea, others expressed concerns about the potential risks. Participant H worried that VR/MR might hinder emotional healing, sharing, *"If it feels too real, I'm afraid I won't be able to move on. It might trap me in the past rather than helping me cope."* Similarly, Participant E added, *"I worry that seeing my dog in VR would make me miss her even more, and I'm not sure if it would help or hurt in the long run."* These concerns reflected fears of becoming overly reliant on virtual representations, especially among those who had already spent significant time processing their grief.

Preferences for Passive vs. Active Interactions Participants expressed a notable preference for passive interactions over active engagement in VR/MR settings. For many, minimal interaction felt more comforting and reflective. Participant J noted, *"I wouldn't need to interact much; just seeing him lying there peacefully or hearing familiar sounds would be enough for me."* This preference for passive experiences may stem from the emotional intensity of grief, as it allows for reflection without overwhelming emotions. However, for those with unresolved grief, active interactions such as playing or talking with a virtual pet were viewed as valuable ways to process emotions and maintain continuing bonds.

4 Design Insights from Preliminary Findings

Our study highlights key design considerations for MR systems supporting pet bereavement, focusing on immersive, memory-rich experiences that balance emotional sensitivity and user engagement. The advantages of MR, such as its ability to blend real-world and virtual elements, directly address the limitations of static grieving methods[24, 32, 37]. This section emphasizes customization, interaction levels, avatar realism, and shared spaces for healing.

4.1 Customization

Every pet-owner relationship is unique, shaped by shared habits and memories[31]. Participants described varied ways of remembering their pets, highlighting the need for personalized MR environments. Prior work demonstrates that customization enhances emotional engagement in bereavement systems[19, 45]. For example, one participant created a memory box with their dog's favorite items, a concept adaptable to MR as "virtual memory boxes" containing representations of favorite toys or treats. These boxes could link to specific photos or videos, creating an immersive connection between virtual objects and real-life memories. Personalization fosters

deeper emotional engagement, making the virtual experience more meaningful.

4.2 Low-Interactive Level Engagement

Many participants expressed a preference for simple and low-intensity interactions with their virtual pets, such as observing them in a peaceful setting or engaging in minimal activities (e.g., using a laser pointer for a cat or tossing a toy for a dog). Participants emphasized that overly realistic or highly interactive scenarios were seen as potentially overwhelming. Research highlights that even low-interaction activities provide meaningful emotional connections [8, 11, 15]. To address these preferences, MR systems should offer low-interactive engagement modes focusing on companionship rather than active play. Figure 2 illustrates a spectrum of interaction levels, ranging from passive observation to active collaboration. For instance, a virtual pet avatar could sit calmly next to the user or engage in simple, predictable activities, offering comfort without emotional overwhelm. These low-interaction features align with grief research suggesting that less active engagement reduces emotional strain [36]. While some participants preferred higher engagement, such as interactive play, most favored simpler interactions, emphasizing the need for customizable engagement levels to accommodate diverse emotional needs.

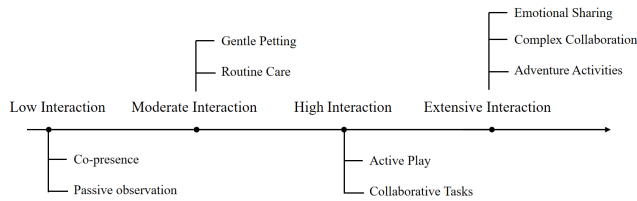


Figure 2: Spectrum of pet-owner interaction levels, ranging from low to high engagement.

4.3 Avoiding Overly Realistic Avatars

While many participants expressed interest in seeing their pets again in MR, they also highlighted the potential emotional challenges of representations that feel too real, such as those created directly from pets' daily videos. Such realistic representations could blur the line between reality and virtuality, intensifying sadness or preventing emotional closure. Research [33] suggests that balancing realism with abstraction is critical in virtual environments to avoid overwhelming emotional responses. Incorporating subtle abstract elements, such as translucent effects, soft halos, or stylistic adaptations, allows users to differentiate between virtuality and reality, offering emotional safety while maintaining a comforting connection.

To address these concerns, pet avatars should incorporate abstraction to provide emotional support without overwhelming users. Figure 3A illustrates three conceptual designs for virtual pet avatars:

- **Realistic Representation:** A lifelike avatar generated from the pet's daily videos, closely resembling the pet's original appearance. While it captures the emotional essence of the



Figure 3: Conceptual designs for MR systems supporting pet bereavement: (A) Three virtual pet avatar designs: a realistic avatar generated from daily videos for familiarity, an abstract representation with angelic wings and flowing motion symbolizing the pet's passing, and a cartoon-style avatar with playful traits to reduce emotional intensity. (B) A serene "pet heaven" offering a visual narrative of harmony and closure, where pet avatars play together in a peaceful environment. (C) A shared memorial space featuring an interactive community message board, allowing users to leave letters for their pets and engage with messages from others, fostering mutual support and connection. *Note: AI-generated for illustrative purposes.*

pet, it risks blurring the line between virtuality and reality, potentially overwhelming users emotionally.

- **Avatar with Wings and Flowing Motion:** A slightly abstract representation where the pet has angelic wings and subtle flowing movements, symbolizing that the pet is no longer alive. This approach conveys a comforting, other-worldly presence while reminding users of the virtual nature of the experience.
- **Cartoon-Style Avatar:** A stylized representation that incorporates key features of the original pet but distances itself from realism. This cartoon-like avatar retains the pet's familiar traits (e.g., fur patterns, size) while presenting a cute, playful appearance, avoiding emotional intensity caused by hyper-realistic visuals.

Additionally, emotional sensitivity features like user control (e.g., pausing or ending the interaction) should be incorporated, allowing users to regulate their experience according to their emotional state. Previous research on grief in MR also supports these strategies, highlighting the need for balance between immersion and emotional safety [23].

4.4 Shared Spaces for Healing and Connection

Participants frequently expressed a desire to create a "happy ending" for their pets, particularly those lost suddenly, imagining them in peaceful, joyful settings to ease unresolved emotions. This highlights that MR systems could offer shared spaces where users can collectively honor their pets. For example, a virtual "pet heaven" could depict pet avatars playing together in a serene environment, providing a visual narrative of harmony and closure (Figure 3B). This aligns with narrative psychology, which suggests that storytelling helps individuals make sense of life events, transforming pain into meaning and resolution [25]. For pet owners with sudden losses, uplifting scenarios like this can reduce distress and promote well-being [14].

Shared memorial spaces could allow users to leave messages, such as letters to their pets, which others can read and respond to, fostering connection and mutual support (Figure 3C). This design

aligns with the CB theory, emphasizing symbolic connections with the deceased to aid coping [9]. By interacting with their pet's avatar and sharing messages in a supportive community, users can maintain emotional ties while processing their grief. The collaborative nature of such spaces fosters belonging, countering the isolation often associated with disenfranchised grief [4, 6].

5 Conclusion

This paper explores how MR systems can support pet bereavement by addressing the emotional needs of grieving pet owners. Through interviews, we identified key design insights, such as the preference for low-interaction pet avatars and the value of abstract representations, to guide the development of MR systems for grief support. These findings provide a foundation for future research, including prototyping and testing MR systems to evaluate usability and emotional impact, as well as investigating cultural and regional factors for inclusivity. Future work will also explore AI-driven personalization and multisensory stimuli, such as haptic feedback and auditory cues, to create immersive, emotionally supportive experiences. These efforts aim to expand the potential of XR technologies in grief support, providing meaningful and adaptive solutions for grieving pet owners.

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